

Unit 1: Introduction to Web Essentials and HTML

Clients, Servers, and Communication, The Internet-Basic Internet Protocols -The World Wide Web, HTTP request message-response message, Web Clients Web Servers-Case Study, Markup Languages: XHTML. An Introduction to HTML History-Versions-Basic XHTML Syntax and Semantics-Some Fundamental HTML Elements-Relative, The and<div> tags ,Lists, Tables.

INTRODUCTION TO WEB PROGRAMMING

Web programming is the process of creating applications and services that run on the Internet or on an intranet (private network). These applications are called **web applications** and are accessed using web browsers such as Chrome, Edge, or Firefox.

Unlike traditional software installed on a single computer, web applications are hosted on servers and can be accessed from anywhere using the Internet.

Web programming combines multiple technologies such as:

- HTML – Structure of web pages
- CSS – Styling and layout
- JavaScript – Interactivity and dynamic behavior
- Server-side languages – Data processing and backend logic

Web programming enables the development of modern systems like online banking, e-commerce websites, social media platforms, and online learning portals.

Evolution of Internet

The evolution of web programming is closely linked with the evolution of the Internet itself. Before web programming and web applications existed, the foundation was laid by early computer networks developed for research and defense purposes. Over time, these networks evolved into the modern Internet, which later enabled the development of web technologies and web programming.

The early development of the Internet began with research funded by the **U.S. Department of Defense**, which aimed to create a reliable communication network capable of functioning even during emergencies such as war or system failures. The primary goals were to improve communication among researchers, enable data and program sharing, and allow remote computer access for defense research. A key design principle was **network robustness**, ensuring the network would continue working even if some nodes failed, with no single point of failure.

This research led to the creation of **ARPANET** in the late 1960s, considered the origin of the modern Internet. ARPANET introduced packet switching technology, where data is divided into small packets and sent through multiple paths to reach the destination efficiently. It enabled email communication, connected research institutions, and laid the foundation for TCP/IP protocols, proving that large-scale computer networks were practical and effective.

During the 1980s, networking expanded beyond military research with the emergence of academic networks such as **BITNET** and **CSNET**. BITNET mainly supported email and file transfer between universities, while CSNET provided networking access to computer science departments that were not connected to ARPANET. These networks played a major role in spreading networking technology into education and research communities.

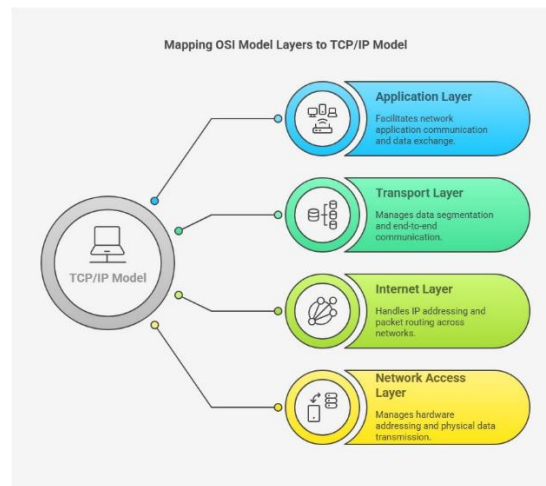
In 1986, the **National Science Foundation** launched **NSFNET**, which became the backbone of the early Internet. NSFNET connected universities and research centers across the United States, provided high-speed networking, supported research collaboration, and enabled large-scale academic data sharing.

By the late 1980s and early 1990s, NSFNET expanded rapidly, increasing network speed and user access. Eventually, it transitioned from a government-funded research network to a commercially supported Internet infrastructure. This shift allowed private Internet Service Providers (ISPs) to emerge, leading to the public and commercial Internet we use today.

What is the Internet?

The Internet is a global system of interconnected computer networks that communicate using standardized protocols. It allows computers, servers, mobile devices, and other digital systems across the world to exchange information. The Internet supports multiple services such as email, web browsing, file transfer, video conferencing, and cloud computing. It acts as the backbone for modern digital communication and web applications. The Internet is often called a “**network of networks**” because it connects millions of smaller private, public, academic, business, and government networks into one global communication system.

TCP/IP Protocol: TCP/IP (Transmission Control Protocol / Internet Protocol) is the fundamental communication protocol suite used in the Internet. It defines how data is transmitted, addressed, routed, and received across networks.



Main Functions

- TCP ensures reliable data transmission
- IP handles addressing and routing
- Breaks data into packets
- Reassembles packets at destination

Higher-Level Protocols: Higher-level protocols operate above TCP/IP and provide specific services for users and applications.

Examples

- HTTP – Web communication
- HTTPS – Secure web communication
- FTP – File transfer
- SMTP – Simple Mail Transfer Protocol

Unique Identification – Internet Protocol (IP) Addresses: Every device connected to the Internet must have a unique IP address. An IP address identifies a device so that data can be sent to the correct destination. IP addresses are numerical labels assigned to each device connected to a network.

Types of IP Addresses

1. IPv4 – 32-bit address (Example: 192.168.1.1)
2. IPv6 – 128-bit address (Used for future expansion)

IPv4 Structure

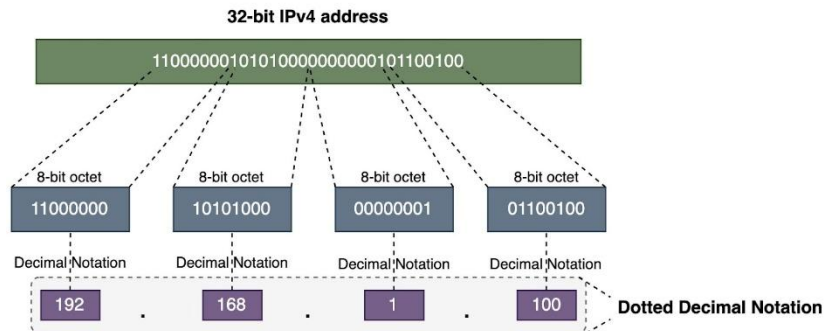
- Divided into four parts (octets)
- Each part ranges from 0 to 255

IP Address Allocation: IP addresses are allocated in a hierarchical manner to ensure uniqueness across the world.

Allocation Process: Global level allocation by Internet governing bodies. Allocation to ISPs (Internet Service Providers). ISPs assign IP addresses to users

Types of Allocation

- Static IP – Fixed address
- Dynamic IP – Changes periodically



DOMAIN NAMES AND IP ADDRESSES

Since IP addresses are difficult to remember, domain names are used as user-friendly names for websites.

Example:

- Domain Name → google.com
- IP Address → Numeric address mapped to domain

Domain names are translated into IP addresses using DNS (Domain Name System).

Domain Name Conversion (DNS Working)

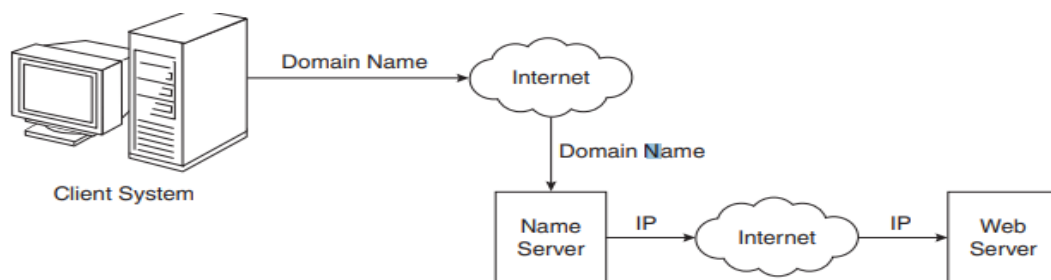


Figure 1 Domain name conversion

The diagram illustrates how a domain name is converted into an IP address so that a client system can communicate with a web server over the Internet. This process is handled by the Domain Name System (DNS). Since humans find it easy to remember names (like websites), but computers communicate using numbers (IP addresses), DNS acts as a translator between them.

Main Components in the Diagram

- 1. Client System:** This is the user's computer or device (browser). It sends a request using a **domain name** (example: www.website.com).
- 2. Internet:** Acts as the communication medium that connects the client system, name server, and web server.
- 3. Name Server (DNS Server):** Stores mappings of domain names to IP addresses. It helps in converting domain names into IP addresses.
- 4. Web Server:** Hosts the actual website content. It responds only when contacted using the correct IP address.

Step-by-Step Working

Step 1 – Client Sends Domain Name Request: The client system sends a domain name request through the Internet.

Example: Client requests → www.example.com

At this stage: Client knows only domain name, does not know IP address

Step 2 – Request Reaches Name Server: The domain name is sent to the Name Server (DNS server). The Name Server: Searches its database. Finds matching IP address

Example: www.example.com → 192.168.X.X

Step 3 – Name Server Sends IP Address: The Name Server sends the corresponding IP address back through the Internet. Now the client knows: Exact location of web server. How to communicate with it.

Step 4 – Client Contacts Web Server Using IP: The client now sends request directly to the web server using the IP address.

The web server then: Processes request, Sends webpage data back to client.

Internet Protocols

Internet protocols are standardized rules that define how data is transmitted, formatted, and received over the Internet. These protocols ensure that different types of computers and networks can communicate with each other efficiently. The Internet mainly works using the **TCP/IP protocol suite**, which allows reliable communication across global networks.

Internet protocols control: Data transmission, Addressing and routing, Error detection and correction, Data formatting

Common Internet Protocols

- **IP (Internet Protocol)** – Handles addressing and routing
- **TCP (Transmission Control Protocol)** – Ensures reliable delivery
- **HTTP/HTTPS** – Web communication
- **FTP** – File transfer
- **SMTP, POP3, IMAP** – Email communication

These protocols work together to support Internet services such as web browsing, email, and file transfer.

Limitations of Early Internet Protocols

Early Internet protocols were designed mainly for research and academic communication, not for commercial or global public use. As Internet usage increased, several limitations were observed.

Major Limitations

- Limited security features
- No encryption in early protocols
- Limited multimedia support
- Not designed for large-scale commercial use
- Limited bandwidth efficiency
- No built-in authentication mechanisms

For example:

- Early HTTP transmitted data in plain text
- Early networks were not designed for millions of users

These limitations led to the development of improved protocols such as HTTPS and modern network security technologies.

The Birth of the World Wide Web: The World Wide Web was developed to make Internet information easy to access and share. Before the Web, the Internet was mainly used by researchers using command-based tools. The Web introduced: Graphical interface, Hyperlinks for navigation, Easy access to information. The Web made the Internet popular among the general public and businesses.

INTRODUCTION TO THE WORLD WIDE WEB

The **World Wide Web (WWW)** is an information system that allows users to access documents and resources through the Internet using web browsers. It is based on technologies such as HTML, HTTP, and URLs.

The Web allows:

- Easy navigation between documents
- Access to multimedia content
- Global information sharing

The Web is only one service that runs on the Internet but is the most widely used one.

Core Concepts/Terminology of the World Wide Web

Hypertext: Hypertext refers to text that contains embedded links (hyperlinks) to other documents or resources. These links allow users to navigate between web documents in a non-linear or non-sequential manner.

Example: Clicking a link in a webpage to move to another webpage.

Importance: Enables easy navigation, Connects related information, Forms the foundation of the Web

Hypermedia: Hypermedia extends hypertext by including multimedia elements such as images, audio, and video along with text.

Includes: Text, Images, Audio, Video, Animations. Hypermedia makes web content more interactive and engaging.

Web Browsers: Web browsers are software applications used to access, retrieve, and display web documents from web servers.

Functions of Browsers: Send requests to web servers, Interpret HTML, CSS, JavaScript, Display web content

Examples: Chrome, Edge, Firefox.

Web Servers: Web servers are systems that store, process, and deliver web content to clients (browsers).

Functions of Web Servers: Store website files, Process client requests, Send responses (web pages)

The Web: The term **Web** is short for **World Wide Web**. It refers to the collection of interlinked web resources accessible through the Internet.

Documents vs Pages vs Resources

These terms are often used interchangeably but have slightly different meanings.

Documents: Technically correct term, usually refers to text-based content, Used in early web terminology.

Pages: Most commonly used term, refers to what users see in browser, Slightly misleading because modern content is more than pages

Resources: Most accurate modern term. Includes all types of content: Text, Images, Videos, Audio, Files, APIs

Web vs Internet

Many people use the terms Internet and Web interchangeably, but they are different. The Internet is the global network infrastructure that connects computers worldwide, while the Web (World Wide Web) is a service that operates on top of the Internet and allows users to access websites and web resources.

Internet : The Internet is a global network of interconnected computers and devices. It provides the physical and logical infrastructure required for communication. It includes networking hardware like routers, cables, satellites, and communication protocols such as TCP/IP.

The Internet supports many services, including: Email, File transfer (FTP), Video conferencing, Online gaming, Web (WWW).

Web / World Wide Web: The Web is a system of interlinked web documents and resources that are accessed through the Internet using web browsers. It uses technologies such as HTML, HTTP, and URLs to display information in the form of websites and web applications.

The Web mainly supports: Websites, Web applications, Online portals, Multimedia content.

Difference Between Web and Internet

Feature	Internet	Web (WWW)
Definition	Global network infrastructure	Service running on Internet
Type	Network	Information system
Developed	Earlier	Later
Main Purpose	Connectivity	Information access
Technologies	TCP/IP, routers, cables	HTTP, HTML, browsers
Examples	Email, FTP, VoIP	Websites, Web apps

HTTP (HYPERTEXT TRANSFER PROTOCOL)

HTTP (Hypertext Transfer Protocol) is the standard communication protocol used for transferring data between web browsers (clients) and web servers on the World Wide Web. It defines the rules, message formats, and procedures that allow clients and servers to exchange information.

Whenever a user opens a website, the browser sends an HTTP request to the server, and the server responds with the requested web page or resource. HTTP works at the **Application Layer** of the TCP/IP model.

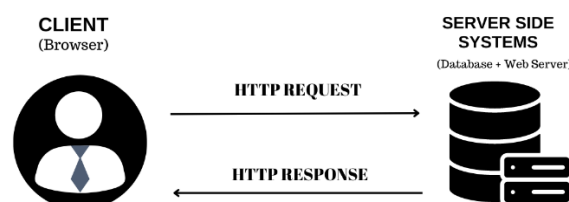
Key Functions of HTTP

- Enables communication between client and server
- Defines request and response message formats
- Transfers web resources like HTML, images, videos, and JSON data
- Supports distributed and scalable web architecture

Stateless Nature of HTTP: HTTP is a stateless protocol, meaning: Each request is treated as a new request. Server does not remember previous client requests. Session tracking requires cookies, tokens, or session IDs.

Example: If you log into a website, HTTP alone does not remember you. Cookies or sessions help maintain login state.

Structure of HTTP Transactions: Every HTTP communication follows a Request-Response Model. This complete exchange is called an HTTP Transaction.



Phase 1: Request Phase: In this phase, the client (web browser) sends a request to the web server to access a resource.

Request Includes: Request Method (GET, POST, PUT, DELETE), URL (resource location), HTTP Version, Headers (metadata), Optional Body (data sent to server)

Component	Meaning	Purpose	Key Points / Examples
Request Method (GET, POST, PUT, DELETE)	Defines action client wants server to perform	Tells server what to do with resource	GET – Retrieve data POST – Send/Create data PUT – Update existing data DELETE – Remove data
URL (Resource Location)	Address of resource on server	Helps server identify which resource is requested	Format: Protocol + Domain + Path + Query Example: https://example.com/products?id=10
HTTP Version	Version of HTTP protocol used	Ensures compatibility between client and server	HTTP/1.0, HTTP/1.1, HTTP/2, HTTP/3
Headers (Metadata)	Extra information about request	Provides data format, authentication, browser details	Examples: Host, Content-Type, User-Agent, Authorization, Cookie
Optional Body	Actual data sent to server	Used to send form data, JSON, files	Mostly used in POST and PUT requests

Comparison of HTTP Request Methods

Parameter	GET	POST	PUT	DELETE
Main Purpose	Retrieve data	Create / Send new data	Update existing data	Delete existing data
Data Location	URL parameters	Request Body	Request Body	URL (mostly)
Security Level	Low	Medium	Medium	Medium
Caching	Yes	No	No	No
Idempotent	Yes	No	Yes	Yes
Common Usage	Fetch webpage / data	Form submission	Update record	Remove record

Sample HTTP Request

GET /index.html HTTP/1.1

Host: www.example.com

User-Agent: Chrome/120.0

Accept: text/html

Phase 2: Response Phase: In this phase, the server processes the request and sends back the response. Response Includes: Status Code (200, 404, 500, etc.), Response Headers, Response Body (actual data)

HTTP Response Structure:

Component	Meaning	Purpose	Example
Status Code	Numerical code showing result	Tells if request was successful or failed	200 OK, 404 Not Found
Status Message	Text description of status code	Helps humans understand result	OK, Not Found
Headers	Metadata about response	Describes content type, length, server info	Content-Type, Date
Body (Optional)	Actual data returned	Sends webpage, JSON data, images	HTML page content

Common HTTP Response Status Code Categories

Code Range	Meaning	Example
1xx	Informational	100 Continue
2xx	Success	200 OK
3xx	Redirection	301 Moved Permanently
4xx	Client Error	404 Not Found
5xx	Server Error	500 Internal Server Error

Sample HTTP Response

HTTP/1.1 200 OK

Content-Type: text/html

Content-Length: 1024

Date: Mon, 16 Feb 2026 10:00:00 GMT

Server: Apache/2.4

ORIGINS AND EVOLUTION OF HTML AND XHTML

INTRODUCTION TO HTML

HTML (Hypertext Markup Language) is a markup language used to structure web documents. It was designed to describe the structure of content such as headings, paragraphs, links, and images. HTML is based on Standard Generalized Markup Language (SGML), an ISO standard for defining markup languages.

Unlike formatting languages, HTML focuses mainly on: Document structure, Logical organization of content, Cross-platform compatibility.

Role of World Wide Web Consortium (W3C) in Standardization

W3C – World Wide Web Consortium

W3C was created to develop standard rules for web technologies so that websites work the same way across different browsers and devices. It released official versions of **HTML (HyperText Markup Language)** such as HTML 2.0, HTML 3.2, HTML 4.0, and HTML 4.01. These standards helped reduce compatibility problems and ensured uniform web development practices worldwide.

Problems with HTML 4.01: HTML 4.01 allowed flexible or loose syntax, which meant developers could write code in different ways, sometimes leading to browser compatibility issues. It also did not define clear error-handling rules, so browsers interpreted incorrect HTML differently. Some outdated (deprecated) features were removed from standards but still supported by browsers, creating confusion.

Transition to XHTML (2000–2001)

XHTML – eXtensible HyperText Markup Language

XML – eXtensible Markup Language

To improve structure and reliability, HTML was rewritten using XML rules, resulting in XHTML. XHTML required strict coding rules such as:

- Proper tag closing
- Case sensitivity
- Proper nesting of elements

XHTML 1.0 allowed gradual transition from HTML, while XHTML 1.1 enforced stricter rules. However, XHTML 2.0 was not widely adopted because it was not backward compatible with existing HTML websites.

Rise of HTML5 and Decline of XHTML 2.0

HTML5 – HyperText Markup Language Version 5

In 2004, the **WHATWG- Web Hypertext Application Technology Working Group** was formed to improve web standards while maintaining compatibility with older HTML versions.

Later, W3C stopped XHTML 2.0 development and worked with WHATWG to develop HTML5. HTML5 became the modern web standard because it supports:

- Multimedia (audio and video)
- Web applications
- Better error handling
- Mobile-friendly web development

This led to the decline of XHTML 2.0.

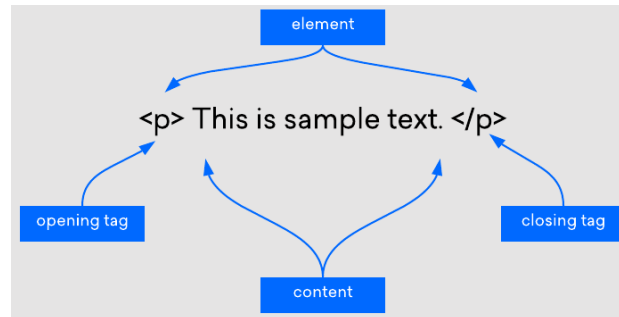
Difference Between HTML and XHTML

Parameter	HTML (HyperText Markup Language)	XHTML (eXtensible HyperText Markup Language)
Definition	Standard markup language for creating web pages	HTML reformulated using XML rules
Syntax Rules	Flexible (lenient)	Strict (must follow XML rules)
Error Handling	Browsers try to fix errors automatically	Errors can break page display
Tag Closing	Optional for some tags	Mandatory for all tags
Case Sensitivity	Not case sensitive	Case sensitive (usually lowercase)
Code Structure	Allows loosely structured code	Requires proper nesting and structure
Compatibility	Works easily across browsers	Requires strict compliance
Ease of Writing	Easier to write and learn	More strict and complex
Current Usage	Widely used (HTML5 today)	Rarely used now

BASIC HTML SYNTAX

HTML (HyperText Markup Language) uses tags to structure web content. A basic HTML document follows a standard structure that tells the browser how to display the content.

HTML TAGS AND ELEMENTS



HTML Tags: HTML tags are keywords enclosed in **angle brackets** `< >`. They tell the browser how to display or structure content.

Types of Tags

- **Opening Tag** → `<p>`
- **Closing Tag** → `</p>`
- **Self-Closing Tag** → `<br>`, `<img>`

Examples

`<p>` → Opening tag
`</p>` → Closing tag
`<img>` → Self-closing tag

HTML Elements: An HTML element is the **complete structure**, including:

- Opening tag
- Content
- Closing tag

Example

`<p>This is a paragraph</p>`

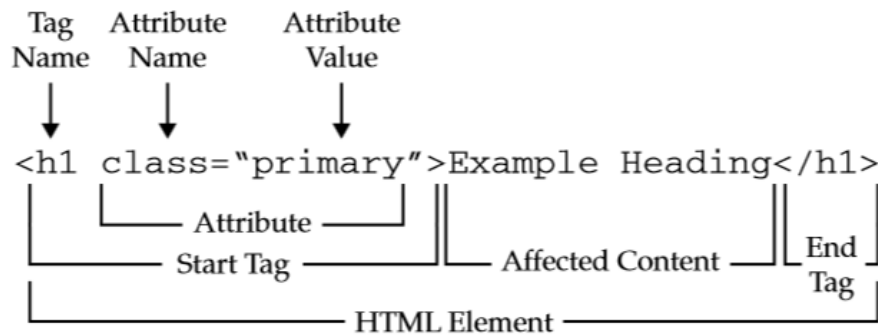
Here:

- `<p>` → Opening tag
- This is a paragraph → Content
- `</p>` → Closing tag

→ Together form **HTML Element**

HTML ATTRIBUTES

A graphical overview of the HTML markup syntax shown so far is presented here:



HTML attributes provide **additional information** about an HTML element. They are always written inside the **opening tag** and usually come in **name = value** pairs.

Attributes help control: Appearance, Behavior, Linking, Data handling.

Basic Syntax

```
<tag attribute="value"> Content </tag>
```

Example

```
<a href="https://www.google.com">Visit Google</a>
```

COMMENTS IN HTML

HTML comments are used to write notes inside HTML code. These notes are not displayed in the browser. Comments help developers understand the code, explain functionality, or temporarily disable code.

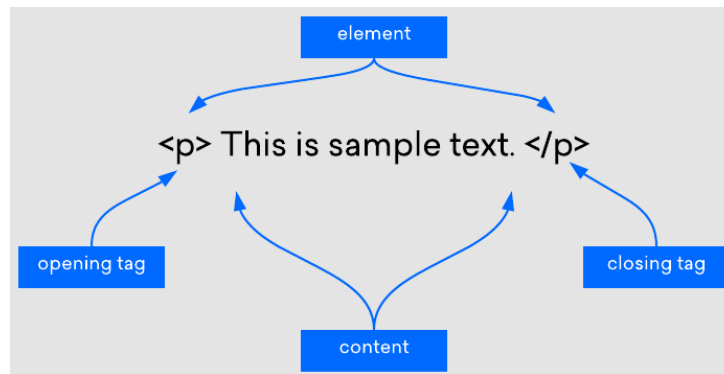
Syntax of HTML Comment

```
<!-- This is a comment -->
```

BASIC TEXT MARKUP IN HTML

Basic text markup in HTML is used to format and structure text content on a webpage. These tags help in organizing content and improving readability by defining headings, paragraphs, and text formatting styles.

<p> Tag in HTML (Paragraph Tag)



The <p> tag is used to define a paragraph in HTML. It is a block-level element that automatically adds space before and after the paragraph, making text easier to read.

Paragraphs are commonly used to display: Text content, Descriptions, Explanations, Article content.

Basic Syntax

<p>Paragraph Content</p>

Simple Example

<p>This is my first paragraph.</p>

```
<!DOCTYPE html>
<html>
<body>

<p>My first paragraph.</p>

</body>
</html>
```

My first paragraph.

Line Breaks –
 Tag in HTML

The
 tag is used to insert a line break in HTML. It forces the text to move to the next line without starting a new paragraph. It is an empty (self-closing) tag, meaning it does not need a closing tag.

Syntax

Example


```
<!DOCTYPE html>
<html>
<body>
```

HTML Code

```
<p>Hello Students<br>
Welcome to Web Programming</p>
```

```
</body>
</html>
```

HTML Code

Hello Students
Welcome to Web Programming

Preserving White Space – <pre> Tag in HTML

The <pre> tag is used to display preformatted text. It preserves: Spaces, Line breaks, Tabs, Indentation. Normally, HTML ignores extra spaces and line breaks, but <pre> keeps them exactly as written in the code.

Basic Syntax

```
<pre>
```

Text content

```
</pre>
```

```
<!DOCTYPE html>
<html>
<body>
```

```
<pre>
Welcome to Web Programming
    This line has spaces
        This line has more
spaces
</pre>

</body>
</html>
```

Welcome to Web Programming
 This line has spaces
 This line has more spaces

HEADINGS (<h1> TO <h6>) IN HTML

HTML headings are used to define titles and subtitles on a webpage. They help organize content and improve readability and SEO (Search Engine Optimization).

1. Types of Headings

HTML provides **6 levels of headings**:

<h1> is biggest by default <h6> is smallest by default

Tag	Purpose
<h1>	Main heading (most important)
<h2>	Subheading under h1
<h3>	Subheading under h2
<h4>	Smaller subheading
<h5>	Less important heading
<h6>	Least important heading

```
<!DOCTYPE html>
<html>
<body>

<h1>Welcome to My Website</h1>
<h2>About Us</h2>
<h3>Our Mission</h3>
<h4>Services</h4>
<h5>Pricing</h5>
<h6>Contact Details</h6>

</body>
</html>
```

Welcome to My Website

About Us

Our Mission

Services

Pricing

Contact Details

BLOCK QUOTATIONS (<blockquote>) IN HTML

The <blockquote> tag is used to define long quotations taken from another source. Browsers usually display block quotes with indentation.

Purpose: Used for large quoted text (paragraph-level quotes). Helps separate quoted content from normal text. Improves semantic meaning (important for accessibility and SEO)

Basic Syntax

<blockquote>

Quoted text goes here...

</blockquote>

```
<!DOCTYPE html>
<html>
<body>

<p>Here is a famous quote:</p>

<blockquote>
The future belongs to those who
believe in the beauty of their
dreams.
</blockquote>

</body>
</html>
```

Here is a famous quote:

The future belongs to those who believe in the beauty of their dreams.

FONT STYLE TAGS

These mainly control **how text looks** (appearance).

Tag	Meaning	Example
	Bold text (no extra importance)	Text
<i>	Italic text	<i>Text</i>
<u>	Underlined text	<u>Text</u>
<small>	Smaller text	Text

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<p>This is <b>bold text</b>.  
</p>
```

```
<p>This is <i>italic text</i>.  
</p>
```

```
<p>This is <u>underlined  
text</u>.</p>
```

```
<p>This is <small>smaller  
text</small>.</p>
```

```
</body>
```

```
</html>
```

This is **bold text**.

This is *italic text*.

This is underlined text.

This is smaller text.

CONTENT-BASED (SEMANTIC) TAGS

These describe the **meaning or importance** of text (not just appearance).

Tag	Meaning	Example
	Important text	Warning
	Emphasized text	Important
<mark>	Highlighted text	<mark>Highlighted</mark>
<code>	Computer code	<code>printf()</code>
<kbd>	Keyboard input	<kbd>Ctrl + C</kbd>
<cite>	Reference title	<cite>Book Name</cite>
<abbr>	Abbreviation	<abbr title="World Health Organization">WHO</abbr>

```
<!DOCTYPE html>
<html>
<body>

<p><strong>Warning:</strong> Do
not turn off the system.</p>

<p>This is <em>very
important</em> information.</p>

<p>This text is
<mark>highlighted</mark> for
attention.</p>

<p>Example of code:
<code>printf("Hello");</code>
</p>

<p>To copy, press <kbd>Ctrl +
C</kbd></p>

<p>Reference:
<cite>Introduction to
Algorithms</cite></p>

<p><abbr title="World Health
Organization">WHO</abbr> works
for global health.</p>

</body>
</html>
```

Warning: Do not turn off the system.

This is *very important* information.

This text is highlighted for attention.

Example of code: `printf("Hello");`

To copy, press ctrl + c

Reference: *Introduction to Algorithms*

WHO works for global health.

SUBSCRIPT AND SUPERScript (<sub> AND <sup>)

Subscript and superscript tags are used to display text slightly below or above the normal text line. These tags are mainly used in scientific formulas, mathematical expressions, and chemical equations. They help represent powers, indices, and chemical compounds correctly in web pages.

Subscript (<sub>): Subscript displays text slightly below the normal text line.

Superscript (<sup>): Superscript displays text slightly **above** the normal text line.

```
<!DOCTYPE html>
<html>
<body>

<p>Water formula: H<sub>2</sub>O</p>
<p>Area formula: x<sup>2</sup></p>

</body>
</html>
```

Water formula: H₂O

Area formula: x²

SPECIAL CHARACTERS (CHARACTER ENTITIES IN HTML)

In HTML, some characters have special meanings. For example, < and > are used for tags, so if we want to display them as normal text, we cannot type them directly. To solve this, HTML uses **Character Entities** (also called **Special Characters**).

Character entities allow us to display reserved symbols, mathematical symbols, currency symbols, and non-keyboard characters correctly in web pages.

General Syntax of Character Entities

&entity_name;

Commonly Used Special Characters

Character	Entity Name	Entity Number	Use
<	<	<	Less than symbol
>	>	>	Greater than symbol
&	&	&	Ampersand
"	"	"	Double quotes
'	'	'	Apostrophe
Space	 	 	Non-breaking space

[illegible]

Special Characters Demo

Display HTML Tags: <p>This is paragraph</p>

Math Symbols: $5 < 10$ and $20 > 15$

Using Ampersand: Tom & Jerry

Quotes: She said "Hello"

Currency: Indian Rupee ₹ 500

Copyright Symbol: © 2026

Extra Space Example: Hello World

HORIZONTAL RULE (<hr>)

The `<hr>` tag in HTML is used to create a **horizontal line (rule)** across a web page. It is mainly used to **separate content sections** and improve readability. It visually indicates a thematic break between topics or sections of content.

In older HTML versions, `<hr>` was mainly used for decoration, but in modern HTML, it represents a **thematic change in content** (for example, change of topic or section).

Basic Syntax

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<p>This is first section</p>
<hr>
<p>This is second section</p>
</body>
</html>
```

This is first section

This is second section

META TAGS (<meta>)

Meta tags in HTML are used to provide **information about the web page** (metadata). This information is not displayed on the webpage but is used by **browsers, search engines, and other web services**.

Meta tags are placed inside the **<head> section** of an HTML document.

Why Meta Tags Are Important

Helps search engines understand the webpage. Defines character encoding. Controls page refresh and redirection. Helps in SEO (Search Engine Optimization). Makes web pages mobile responsive.

Basic Syntax

<meta attribute="value">

<meta> is an **empty tag**. No closing tag required

```
<!DOCTYPE html>
<html>
<head>
<title>Meta Tag Example</title>

<meta charset="UTF-8">
<meta name="description"
content="Learning Meta Tags in HTML">
<meta name="keywords" content="HTML,
Meta Tags, Web">
<meta name="author" content="Student">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">

</head>

<body>
<h2>Meta Tag Example Page</h2>
</body>

</html>
```

Meta Tag Example Page

HTML LISTS: HTML lists are used to organize information in a structured and readable format. Lists help present related items together, making content easier to understand for users and easier to design for developers.

HTML provides **three main types of lists**:

- Ordered List
- Unordered List
- Description (Definition) List

ORDERED LIST (NUMBERED LIST)

An ordered list displays items in a **numbered or ordered sequence**.

 → Ordered List

 → List Item

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>

<body>
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>

</body>

</html>
```

1. HTML
2. CSS
3. JavaScript

Ordered List Types

Type	Output Style
type="1"	1, 2, 3
type="A"	A, B, C
type="a"	a, b, c
type="I"	I, II, III
type="i"	i, ii, iii

UNORDERED LIST (BULLETED LIST): An unordered list displays items using bullets instead of numbers.

`<ul>` → Unordered List

`<li>` → List Item

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>

<body>
<ul>
  <li>Mouse</li>
  <li>Keyboard</li>
  <li>Monitor</li>
</ul>

</body>

</html>
```

- Mouse
- Keyboard
- Monitor

Bullet Types

Type	Style
Disc	• (default)
Circle	○
Square	■

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>

<body>
<ul type="square">
  <li>Pen</li>
  <li>Pencil</li>
</ul>

</body>

</html>
```

- Pen
- Pencil

DESCRIPTION LIST (DEFINITION LIST): Used to display **terms and their descriptions**.

`<dl>` → Description List `<dt>` → Description Term `<dd>` → Description Description

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>

<body>
<dl>
  <dt>HTML</dt>
  <dd>Used to create web page
structure</dd>

  <dt>CSS</dt>
  <dd>Used to style web pages</dd>
</dl>

</body>
</html>

```

HTML

Used to create web page structure

CSS

Used to style web pages

Combined example:

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<h3>Web Technologies</h3>

<ol>
  <li>Frontend</li>
  <li>Backend</li>
</ol>

<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<dl>
  <dt>HTML</dt>
  <dd>Structure of webpage</dd>
</dl>
|
</body>
</html>

```

Web Technologies

1. Frontend
2. Backend

- HTML
- CSS

HTML

Structure of webpage

Nested Lists in HTML: A Nested List means placing one list inside another list. It is used when we want to show sub-items under a main item. Nested lists help in representing hierarchical or multi-level data clearly.

Example 1 – Nested Unordered List

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<h3>Web Technologies</h3>

<ul>
  <li>Programming Languages
    <ul>
      <li>Python</li>
      <li>Java</li>
      <li>C++</li>
    </ul>
  </li>

  <li>Web Technologies
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>
  </li>
</ul>

</body>
</html>
```

Web Technologies

- Programming Languages
 - Python
 - Java
 - C++
- Web Technologies
 - HTML
 - CSS
 - JavaScript

Example 2 – Ordered List with Nested Unordered List

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<h3>Web Technologies</h3>

<ol>
  <li>Frontend
    <ul>
      <li>HTML</li>
      <li>CSS</li>
    </ul>
  </li>

  <li>Backend
    <ul>
      <li>Node.js</li>
      <li>Python</li>
    </ul>
  </li>
</ol>

</body>
</html>

```

Web Technologies

1. Frontend
 - HTML
 - CSS
2. Backend
 - Node.js
 - Python

Example 3 – Multi-Level Nested List

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<h3>Web Technologies</h3>

<ul>
  <li>Engineering
    <ul>
      <li>Computer Science
        <ul>
          <li>AI</li>
          <li>Data Science</li>
        </ul>
      </li>
    </ul>
  </li>
</ul>

</body>
</html>

```

Web Technologies

- Engineering
 - Computer Science
 - AI
 - Data Science

INTRODUCTION TO TABLES IN HTML

HTML tables are used to **organize and display data in rows and columns**. Tables are very useful when we need to present structured data such as student marks, employee details, timetables, and reports.

A table is created using the <table> tag, and it consists of rows, columns, and cells.

Basic Structure of an HTML Table

HTML tables are built using multiple tags working together.

Tag	Meaning
<table>	Creates the table
<tr>	Table row
<th>	Table heading cell
<td>	Table data cell

Basic Table Example

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<table border="1">
  <tr>
    <th>Name</th>
    <th>Course</th>
    <th>Marks</th>
  </tr>

  <tr>
    <td>Ravi</td>
    <td>CSE</td>
    <td>85</td>
  </tr>

  <tr>
    <td>Anita</td>
    <td>ECE</td>
    <td>90</td>
  </tr>
</table>

</body>
</html>
```

Name	Course	Marks
Ravi	CSE	85
Anita	ECE	90

TABLE ATTRIBUTES (BASIC / TRADITIONAL)

Attribute	Purpose
border	Adds border to table
cellpadding	Space inside cell
cellspacing	Space between cells
width	Table width

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<table border="2"
        cellpadding="10"
        cellspacing="5"
        width="60%"
        align="center"
        bgcolor="#E6F2FF"
        bordercolor="blue">

<tr bgcolor="#99CCFF">
  <th>Name</th>
  <th>Department</th>
  <th>Marks</th>
</tr>

<tr align="center">
  <td>Rahul</td>
  <td>CSE</td>
  <td>85</td>
</tr>

<tr align="center">
  <td>Sneha</td>
  <td>ECE</td>
  <td>90</td>
</tr>

</table>
</body>
</html>
```

Name	Department	Marks
Rahul	CSE	85
Sneha	ECE	90

THE ROWSPAN AND COLSPAN ATTRIBUTES IN HTML TABLES

In HTML tables, rowspan and colspan are used to **merge table cells**. These attributes help create complex table layouts by allowing one cell to span across multiple rows or columns.

1. rowspan Attribute: The rowspan attribute is used to make a cell span (extend) across **multiple rows**.

Syntax

```
<td rowspan="number">
```

The number indicates how many rows the cell should cover.

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<table border="1">
<tr>
<th>Name</th>
<th>Subject</th>
</tr>

<tr>
<td rowspan="2">Rahul</td>
<td>Maths</td>
</tr>

<tr>
<td>Science</td>
</tr>
</table>

</body>
</html>

```

Name	Subject
Rahul	Maths
	Science

colspan Attribute: The colspan attribute is used to make a cell span across multiple columns.

Syntax

```
<td colspan="number">
```

The number indicates how many columns the cell should cover.

```

<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<table border="1">
<tr>
<th colspan="2">Student Details</th>
</tr>

<tr>
<td>Name</td>
<td>Marks</td>
</tr>
</table>
</body>
</html>

```

Student Details	
Name	Marks

Combined Example – Rowspan + Colspan

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<table border="1">

<tr>
  <th>Name</th>
  <th colspan="2">Marks</th>
</tr>

<tr>
  <td rowspan="2">Anita</td>
  <td>Maths</td>
  <td>90</td>
</tr>

<tr>
  <td>Science</td>
  <td>95</td>
</tr>
</table>
</body>
</html>
```

Name	Marks	
Anita	Maths	90
	Science	95

Table Sections (<thead>, <tbody>, <tfoot>)

HTML provides special tags to divide a table into logical sections. These tags help organize table data clearly and improve readability, accessibility, and styling.

The three main table section tags are:

- <thead> → Table Header Section
- <tbody> → Table Body Section
- <tfoot> → Table Footer Section

1. <thead> – Table Header Section: Contains header rows (column headings). Usually contains <th> tags.

2. <tbody> – Table Body Section: Contains main table data. Usually contains multiple <tr> rows.

<tfoot> – Table Footer Section: Contains summary or totals. Usually appears at bottom visually.

Combined Example

```
<head>
<title>Example</title>
</head>
<body>
<table border="1">
<thead>
<tr>
<th>Student</th>
<th>Marks</th>
</tr>
</thead>

<tbody>
<tr>
<td>Ravi</td>
<td>80</td>
</tr>

<tr>
<td>Anita</td>
<td>90</td>
</tr>
</tbody>

<tfoot>
<tr>
<td>Total</td>
<td>170</td>
</tr>
</tfoot>
</table>
</body>
</html>
```

Student	Marks
Ravi	80
Anita	90
Total	170

Problems with Using Tables for Page Layout

Earlier, tables were used to design web page layouts, but this caused many issues:

Too many unnecessary tags – Tags like <table>, <tr>, <td> made HTML very long and messy.

Poor performance – Large HTML files increased page loading time.

Difficult maintenance – Changing design required editing many nested tables.

High complexity – Developers even used invisible images just for alignment.

Industry Shift After Dotcom Bubble (2001): After 2001, web development practices improved:

More skilled and trained developers entered the field.

Knowledge and usage of CSS increased.

New trend started → **Tableless Layout (Using CSS for design instead of tables).**

Why CSS is Better Than Tables for Layout

Faster Page Load – CSS keeps HTML smaller and cleaner.

More Flexibility – Easy to create responsive designs for mobile, tablet, and desktop.

Better Accessibility – Screen readers and search engines understand structured HTML + CSS better.

Easy Maintenance – Design changes can be done in one CSS file instead of editing many HTML pages.

 and <div> Tags in HTML

 and <div> are **container tags** used to group and organize content in HTML. They are mainly used with **CSS** and **JavaScript** for styling and layout purposes.

Both tags do not add any visual change by default. They are used to structure content logically.

1. <div> Tag (Block-Level Container)

The <div> tag is a **block-level element**. It takes the full width of the page and starts on a new line.

Uses of <div>

- ✓ Group large sections of content
- ✓ Create page layout (header, footer, sidebar)
- ✓ Apply CSS styles to sections
- ✓ Used in modern web design

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<div style="background-color:
lightblue;">
  <h2>Welcome</h2>
  <p>This is inside div</p>
</div>

</body>
</html>
```

Welcome

This is inside div

 Tag (Inline Container)

The tag is an **inline element**. It only takes the space required and does not start on a new line.

Uses of

- ✓ Style part of text
- ✓ Apply color or font to specific words
- ✓ Used inside paragraphs

Key Differences Between <div> and

Feature	<div>	
Type	Block-level	Inline
Line Break	Starts new line	Same line
Width	Full width	Only needed width
Used For	Large sections	Small text parts

Combined Example

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<div style="border:1px solid black;
padding:10px;">
  <h3>Student Info</h3>
  <p>Name: <span
style="color:blue;">Rahul</span></p>
</div>
</body>
</html>
```

Student Info

Name: Rahul

```
<!DOCTYPE html>
<html>
<head>
<title>Example</title>
</head>
<body>
<p>This is <span
style="color:red;">important</span> text.</p>
</body>
</html>
```

This is important text.